



Additional Tips for Improving Models

You just learned some important tips for addressing common issues with object detection models. Here are a few additional suggestions.

1. Revisit the ground truth. Load the ground truth into the Image or Video Labeler App and check that it meets your expectations. Was the labeling consistent? Is it correct? Fixing errors and inconsistencies may help, especially with smaller datasets.

2. Try another backbone and/or detection framework. A good place to start is using YOLOX with the "tiny-coco" backbone because you can train and test models more quickly. If you are not achieving the desired results, try using the "small-coco" backbone or switch to YOLO v4 with the "csp-darknet53-coco" backbone. The larger backbones can handle more complexity in the data. The tradeoff is training time and prediction speed.

3. Evaluate your model using several IoU thresholds. Your application determines the required location accuracy. However, evaluating your model with a small range of IoU thresholds centered around your requirement will tell you how sensitive your model is to the IoU threshold.

If your AP and mAP change significantly with IoU, then there's likely significant variation in the size and orientation of your objects that is not well captured in your data. Acquire more data or use data augmentation, which you'll learn about in course 3.

4. Examine how your mAP vs Detection Threshold curve decreases. If your mAP is satisfactory at low detection thresholds but rapidly decreases with increasing detection threshold, examine the AP curve for each class and look for classes with true positives after many false positives. Also, look for classes with low recall.

True positives after false positives mean there are correct detections with lower confidence than false positives. Low recall means the detector is missing many objects regardless of the detection threshold.

Acquire more data for the low-performing classes or use data augmentation to add more variation.

5. Acquire more data. Training accurate models often requires lots of data. Collecting and labeling images is a time-consuming task. In course 3, you'll use AI-assisted labeling to help label new images and use data augmentation to generate synthetic images.

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